Name and formula

Reference code:	04-010-5696
Mineral name:	Moissanite-2H, syn
Compound name:	Silicon Carbide
Empirical formula:	CSi
Chemical formula:	SiC
Mineral classification:	Wurtzite (Supergroup), 2H (Group)

Crystallographic parameters

Crystal system:	Hexagonal
Space group:	P63mc
Space group number:	186
a (Å):	3.0815
b (Å):	3.0815
c (Å):	5.0307
Alpha (°):	90.0000
Beta (°):	90.0000
Gamma (°):	120.0000
Calculated density (g/cm^3):	3.22
Volume of cell (10^6 pm^3):	41.37
Z:	2.00
RIR:	1.78

Subfiles and quality

Subfiles: Alloy, metal or intermetalic, Ceramic, Ceramic - Semiconductor, Forensic, Inorganic, LPF Pattern, Mineral, Mineral - Mineral, Mineral - Synthetic

Quality: Indexed (I)

Comments

ANX:	NO
Creation Date:	9/1/2007

Modification Date: 9/1/2011
Cross-References: LPF:1411639, ICDD:04-010-5696
ANX: NO
LPF Collection Code: 1411639
Sample Preparation: STARTING MATERIALS: SiC Compound Formation: hot-pressing at 1873-2373 K. Article Title: PHASE TRANSFORMATION DURING HOT-PRESSING OF CUBIC SiC. Minor Warning: No R factors reported/abstracted. No e.s.d reported/abstracted on the cell dimension. Wyckoff Sequence: b2. Unit Cell Data Source: Powder Diffraction.

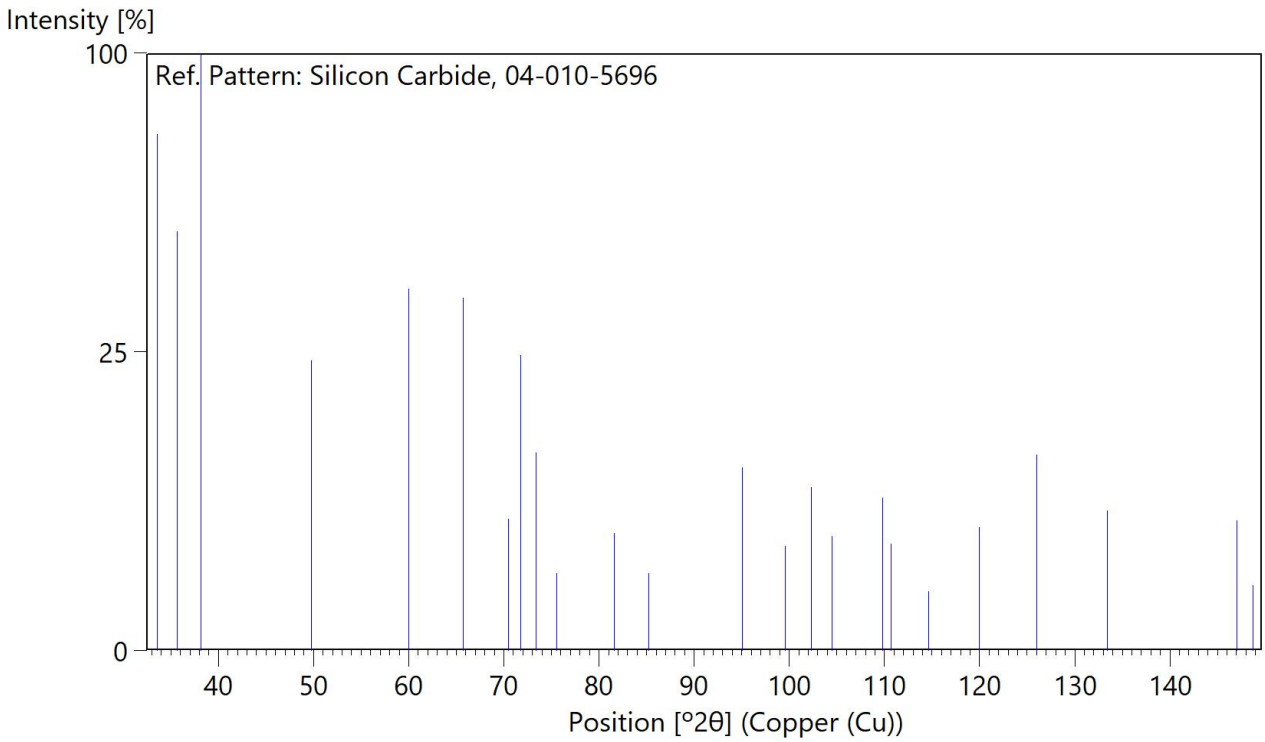
References

Primary reference: Bind, J., *Mater. Res. Bull.*, **13**, 91-96, (1978)
Structure: Bind, J., *Mater. Res. Bull.*, **13**, 91-96, (1978)

Peak list

No.	h	k	l	d [Å]	2θ [°]	I [%]
1	1	0	0	2.66866	33.554	74.9
2	0	0	2	2.51535	35.665	49.4
3	1	0	1	2.35749	38.143	100.0
4	1	0	2	1.83042	49.774	23.7
5	1	1	0	1.54075	59.993	36.7
6	1	0	3	1.41985	65.711	35.0
7	2	0	0	1.33433	70.521	4.9
8	1	1	2	1.31386	71.788	24.5
9	2	0	1	1.28973	73.347	11.0
10	0	0	4	1.25768	75.538	1.7
11	2	0	2	1.17875	81.610	3.9
12	1	0	4	1.13767	85.233	1.7
13	2	0	3	1.04412	95.080	9.4
14	2	1	0	1.00866	99.580	3.1
15	2	1	1	0.98897	102.317	7.5
16	1	1	4	0.97430	104.487	3.7
17	1	0	5	0.94145	109.811	6.6
18	2	1	2	0.93619	110.732	3.2
19	2	0	4	0.91521	114.632	0.8
20	3	0	0	0.88955	119.981	4.3
21	2	1	3	0.86434	126.048	10.8
22	3	0	2	0.83865	133.414	5.5
23	0	0	6	0.83865	133.414	5.5
24	2	0	5	0.80335	147.015	4.8
25	1	0	6	0.79990	148.729	1.2

Structure

No.	Name	Element	X	Y	Z	Biso	sof	Wyck.
1	SI1	Si	0.66666	0.33333	0.50000	0.5000	1.0000	2b
2	C1	C	0.66666	0.33333	0.87500	0.5000	1.0000	2b

Stick Pattern